

Davide **Donati**

Quantum Engineering Student

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Profile

Master's student in Quantum Engineering at Politecnico di Torino with a Bachelor's degree in Computer Science. My main interests are quantum computing and quantum machine learning, supported by hands-on experience with quantum circuits (**Qiskit**, **PennyLane**, **Quirk**), real quantum hardware (**IQM**), quantum network simulation (**NetSquid**), qubit control electronics (**FPGA**, **RFSoC**) and high-performance computing (**OpenMP**, **MPI**, **CUDA**).

Education

Politecnico di Torino

Turin, Italy

Master's Degree in Quantum Engineering — *Sep. 2025 – Present*

- Particular focus on quantum computing and its applications.
- Executed quantum circuits on the real **IQM** quantum computer available at Politecnico di Torino.
- Laboratory work on qubit electronics and experiments on the university HPC cluster (see *Quantum and Research Projects*).

University of Bologna

Bologna, Italy

Bachelor's Degree in Computer Science — *Sep. 2021 – Mar. 2025*

- Thesis: classical vs. quantum machine learning for missing data imputation.
- Particular interest in Artificial Intelligence and Quantum Engineering.

Experience

Qubito – Quantum Computing Student Team

Politecnico di Torino, Italy

Member — *Sep. 2025 – Present*

- Participation in quantum computing hackathons.
- Research on quantum technologies and science outreach.

PQID-Bench

Quantum Code Reviewer — *Sep. – Oct. 2026*

- Reviewed AI-generated **Qiskit** code to assess whether it was correct and physically plausible, contributing to the construction of the PQID-Bench benchmark.

Serie A

Italy

Professional Volleyball Player — *2022 – 2024*

- Competed at top level while completing my Bachelor's degree (Lega Pallavolo Profile).

Quantum and Research Projects

Qubit Electronics Lab

Politecnico di Torino — *2026*

- Work on **FPGAs** and use of **RFSoC** platforms.
- Simulation of quantum dots, their charge and behavior.
- Analysis of basic electronic components and circuits.

Quantum Network Simulation

Politecnico di Torino — *2026*

- Simulation of an **entanglement swapping** protocol with **NetSquid**.

High-Performance Computing Project

Politecnico di Torino — *2026*

- Experiments on the Politecnico HPC cluster using **OpenMP**, **MPI** and **CUDA** to simulate waves and to stress-test the HPC system itself.

Quantum Machine Learning for Missing Data Imputation

Bachelor's Thesis, University of Bologna *2025*

- Analysis of the differences between classical and quantum machine learning for imputing missing data in a dataset.
- Design of a quantum circuit to compare the performance and limits of quantum models against classical ones.

Technical Skills

Quantum Computing	Qiskit, PennyLane (currently preparing for QHack), Quirk, NetSquid, quantum circuit design, quantum ML
Quantum Hardware	IQM quantum computer (hands-on), FPGA, RFSoC, quantum dot simulation, LTspice, basic electronic circuit analysis
HPC	OpenMP, MPI, CUDA on a university cluster; profiling with Intel VTune and NVIDIA Nsight; ParaView
Machine Learning	TensorFlow, Keras (e.g. U-Net for image deblurring)
Programming	Python, C/C++, MATLAB, Java, Haskell, SQL, MongoDB
Tools	Git (GitHub, GitLab), Docker, Jupyter, Microsoft Office
Other	Web development (MERN, HTML/CSS/JS), Unity, Blender, Adobe Suite

Other Projects

All projects below and many more on github.com/TucanoCode04

- **Image Deblur**: U-Net in Keras/TensorFlow reconstructing sharp images from blurred versions.
- **Panda-OS**: operating system with multitasking, memory management and a basic interface.
- **Chesscake**: web app to play chess variants online against players or the computer, with a global leaderboard.
- **Selfie**: web app with calendar, study timer, shared notes and group chat.
- **Programming Game**: endless RPG where players advance through rooms and levels, collecting upgrades.

Languages

Italian	Native speaker
English	C2 – Near-native proficiency
Spanish	B2/C1 – Advanced proficiency
German	B2 – Intermediate proficiency

Interests

Sports	Gym and powerlifting; judo and Brazilian jiu-jitsu; former professional volleyball player
Travel	Lived with host families in Germany, Spain and England; travelled across 14 European countries and overseas
Drive	When I commit to something, I aim for the top: I reached Serie A in volleyball and one of the highest competitive ranks in online strategy games